The world brane of 7+1 spacetime: provenance

The task, the decisions, the inputs, the data, the derivations and the verification record of the document on the world brane

Repository: github.com/once-ere/7plus1 — prepared for Nash
Notebook: 7+1.nb (Wolfram Language 15.0.1)
Document generated by tools/make_docs.py on 2026-10-28

Abstract

This record states what was asked, how it was interpreted, every decision taken (among them the new sections 25 and 26 of 7+1.nb, which compute the worked examples), which results of the earlier sections and which data were used and where they come from, how each result was derived, and how it was checked: the 27 checks of section 25 (27 True) and the 33 of section 26 (33 True), the 37 and 31 boolean checks of the independent Python programs tools/ref_torus_brane.py and tools/ref_radion.py (37 and 31 True), the comparison of the 17 and 15 groups of numbers printed by the notebook and by them, the check that the earlier sections of the notebook are unchanged, the program versions and the SHA-256 hashes of the files.

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1. The task

On 28 October 2026 the author asked for the following (summarized; the original wording is the author's private working material and is not reproduced):

1. Define and explain the "world brane" precisely and correctly, in its own documentation file with its own symbol file and provenance file, each in Markdown and LaTeX with the LaTeX compiled to PDF; explain it for students, clearly, with examples, well organized; do the same, separately, for the "auxiliary torus"; push the files and any other relevant files to github.com/once-ere/7plus1.

2. Then: read a note of the author on the radion (a private PDF in the workspace, not part of the repository), use its ideas to verify and clarify the radion and brane documents in preparation, and create examples in which the four extra space dimensions are (1) a 4-sphere and (2) a 4-torus. For this document that means: the brane on a 4-sphere compared with the brane on the torus (section 12 of the document), branes in a warped bulk as the note describes them (section 13), and the mixing of the Higgs boson on the brane with the radion (section 7.5).

The standing rules of the project apply: every documentation file has a companion provenance file and a symbol file, in Markdown and LaTeX, compiled to PDF; no file refers the reader to another documentation, provenance or symbol file; every equation that is created or used is written in the file that uses it. The repository's .gitignore is respected.

Interpretation. The world brane is the submanifold $W = \{x_4 = x_5 = x_6 = x_7 = 0\}$ on which 7+1.nb puts the Standard Model (sections 17–24). "Precisely" is read as: a definition by an embedding, the induced metric, the extrinsic curvature, the energy–momentum tensor and the couplings, each stated as an equation; "correctly" as: every worked example is computed and checked twice, by the notebook and by an independent program. The coordinates of 7+1.nb are kept: $X = (x_0, \ldots, x_7)$, x_0 the time (the direction of the octonion unit $e_0 = 1$), x_1, x_2, x_3 ordinary space, $x_4, \ldots, x_7$ the four auxiliary space-like coordinates.

1.1 The signature: the author's explanation of 27 October 2026

Source. On 27 October 2026 the author of the project explained that the xAct list $\{7, 1, 0\}$ declares seven space-like dimensions and one time-like dimension, while $\mathrm{diag}(+1, -1, \ldots, -1)$ is simply the squares of the octonion basis, and asked that this be made clear in all documentation, examples and provenance records. **What it corrected.** The version of 7+1.nb pushed in the commits 52d8729 to dc2f178 of github.com/once-ere/7plus1 declared the xAct metric of section 12 with `DefMetric[{1, 7, 0}, …]`, reading the list as the signs of η; since then section 12 declares `DefMetric[{7, 1, 0}, g71[-a, -b], CD, …]`, exactly as the author's notebook Einstein-Lovelock-7+1-Nash.nb does. **What was added.** Checks in 7+1.nb — section 4.7: η is the squares of the octonion basis, $e_A e_A = \eta_{AA} e_0$ and $\mathrm{Re}(e_A e_B) = \eta_{AB}$; section 12.2: `SignatureOfMetric[g71]` $= \{7, 1, 0\}$ with `SignDetOfMetric[g71]` $= -1$, $\mathrm{Re}(e_A e_B) = \eta_{AB}$, and $\mathrm{Re}(x\,x) = \eta_{AB} x^A x^B$; section 17: the flat metric $\mathrm{diag}(+1, -1, -1, -1)$ of the world brane is the squares of the basis e_0, e_1, e_2, e_3 of the quaternion subalgebra — and the same statements in the Python programs `tools/ref71.py`, `tools/ref_gravity.py` and `tools/ref_sm.py`; the explanation itself in the text of 7+1.nb (sections 1.2, 1.4, 4.2, 4.7, 7, 11, 12.2, 13, 15.1, 17) and in every document, symbol file and provenance file of the repository. **What did not change.** No result. Compared output by output with the backup below, every output of 7+1.nb is the same, except the declared list itself, the new checks and the check counts (and, as in every run, the name of the private context and the measured times); in particular every component of every Lovelock tensor is unchanged. The components follow from the component matrix of the metric, and the sign of the determinant, `SignDetOfMetric`, is -1 for both lists. **Backup.** The evaluated notebook as it was before is kept in the workspace as `backups/7+1_2026-10-27_before_signature.nb` (the folder backups/ is excluded from the repository by .gitignore), 1034401 bytes, SHA-256 `060942488caf4f45120ffd37be0123e96992a31cd7fdbde09ee4dc4ced4f5d3a`; `tools/verify_7p1_nb.py` compares every input, text and section cell of 7+1.nb with it and finds every cell present, each one unchanged or changed by this explanation, and it compares every output in the same way.

The explanation, in full. The 7+1 spacetime has one time-like dimension (x_0, the direction of the octonion unit $e_0 = 1$) and seven space-like dimensions (x_1, x_2, x_3 ordinary space, $x_4, \ldots, x_7$ auxiliary space). Two different objects describe this, and neither may be replaced by the other. (i) The tensor package xAct, used in section 12 of 7+1.nb exactly as in the author's notebook Einstein-Lovelock-7+1-Nash.nb, declares the metric with `DefMetric[{7, 1, 0}, g71[-a, -b], CD, …]`: the list $\{7, 1, 0\}$ declares seven space-like dimensions, one time-like dimension and no degenerate dimension — the numbers of entries $+1$ and -1 of $\eta_{71} = \mathrm{diag}(-1, +1, \ldots, +1)$ (`η71`), the same spacetime written in the sign convention of 4+4.nb (space-like $= +1$, time-like $= -1$). (ii) The matrix $\eta = -\eta_{71} = \mathrm{diag}(+1, -1, \ldots, -1)$ (`η17`) is simply the squares of the octonion basis: $e_0^2 = +1$ for the unit $e_0 = 1$ (time) and $e_i^2 = -1$ for the seven imaginary units $e_1, \ldots, e_7$ (space); more generally it is the real part of the product of two basis units, and it gives the quadratic form of the octonions:

$$\eta_{AB} = \mathrm{Re}(e_A e_B), \qquad e_0^2 = +1, \qquad e_i^2 = -1 \quad (i = 1, \ldots, 7),$$
$$\mathrm{Re}(x\,x) = x_0^2 - x_1^2 - \cdots - x_7^2 = \eta_{AB}\, x^A x^B \qquad (x = x^A e_A \in \mathbb{O}).$$

η gives the signs of the Clifford relation $\Gamma_A \Gamma_B + \Gamma_B \Gamma_A = 2\,\eta_{AB}\, 1_{16}$ of the real 16×16 Dirac matrices: time squares to $+1_{16}$ and each of the seven space directions to -1_{16}, exactly like the octonion units (in the notation $Cl(p, q)$, with p generators squaring to $+1$ and q to -1, the algebra $Cl(1, 7) \cong \mathbb{R}(16)$). It also gives the signs of the metric components: $g_{00} > 0$ along the time-like direction, negative squares along the space-like directions. Every component computed by 7+1.nb follows from the component matrix of the metric; the declared list fixes for xAct the sign of the determinant of the metric, `SignDetOfMetric[g71]` $= -1$, which is the same in both sign conventions, since $\det \eta = \det \eta_{71} = -1$. 7+1.nb checks `SignatureOfMetric[g71]` $= \{7, 1, 0\}$, `SignDetOfMetric[g71]` $= -1$, $\mathrm{Re}(e_A e_B) = \eta_{AB}$ and $\mathrm{Re}(x\,x) = \eta_{AB} x^A x^B$.

2. Decisions

#	decision	reason / equation
1	append section 25 to 7+1.nb (the auxiliary torus and the world brane) and, for the second request, section 26 (the radion, on T^4 and S^4, flat and warped); the summary of all checks becomes section 27	every worked example is computed and checked by the notebook itself, in the same evaluation as everything else
2	write independent Python programs, tools/ref_torus_brane.py and tools/ref_radion.py	their own lattice sums, symbolic integrals, curvature routines and multi-precision arithmetic; no code shared with the notebook
3	back up the evaluated notebook before each appended section and check that the earlier sections are unchanged	tools/verify_7p1_nb.py compares every cell and output with the backups (section 6.3)
4	the brane sits at $\vec{y} = 0$ and is not a dynamical field	as in 7+1.nb; all points of the torus are equivalent, so this is a choice of origin
5	the induced metric $h = J^{\mathsf{T}} g J$; the brane volume element $\sqrt{\lvert h\rvert}$	the definition of a submanifold metric
6	brane energy–momentum $T^W_{\mu\nu} = (2/\sqrt{\lvert h\rvert})\, \delta S_W / \delta h^{\mu\nu}$, with $T^0{}_0$ the energy density	the convention of 7+1.nb; it gives $T^W_{\mu\nu} = f_W h_{\mu\nu}$ for a tension
7	the inhomogeneous metric of section 11.3 is used as the example of a brane that is not totally geodesic	it depends on x_4 and is not even in x_4, so its Christoffel symbols at the brane do not vanish
8	the brane spinors from the Schur average of the 256 Clifford products	Schur's lemma: the intertwiner is unique up to a phase
9	the sphere comparison: brane at the north pole of S^4, the same volume $V_\perp$	the same Newton constant, so that only the geometry differs
10	warped branes in five dimensions (RS1), in the conventions $G^A{}_B - \Lambda \delta^A_B = \kappa T^A{}_B$; tensions from the δ-function part of the field equations	the note's warped formulas are five-dimensional; this form involves no choice of normal
11	the Higgs–radion mixing in the conventions of Giudice, Rattazzi and Wells, translated to the signature $(+,-,-,-)$; the numbers in 60-digit arithmetic	the note follows them; the rate changes, $\sim 10^{-31}$, are below double precision
12	in this document the Higgs field is h and the radion ϕ	the notation of the Standard Model section; $h_{\mu\nu}$ keeps its indices
13	numbers today with $M_8 = 10$ TeV, $M_{\rm Pl} = 2.435 \times 10^{18}$ GeV, $m_r = 10$ GeV, Planck 2018	the choices of sections 22–24

3. Inputs from the earlier sections of 7+1.nb

object	Mathematica	section	equation or value
cosmological and inhomogeneous metrics, curvature	`gCos`, `gPoly`, `cvCos`, `cvPoly`	11	$ds^2 = n^2 dx_0^2 - a^2 d\vec{x}^2 - b^2 d\vec{y}^2$
Lovelock densities	`lovelock71.py`	12	L_2, L_3 of the document
Standard Model, brane representation, intertwiner	`g4D`, `g5D`, `rhoK`, `rBr`, `uInt`, `rho4567`, `PL17`	17	$\Gamma_A U = U R_A$; $g_* = 106.75$; electroweak constants
Kaluza–Klein operator, neutrino portal	`xKK`	18	$X = \gamma_5 \otimes \vec{\rho} \cdot \vec{q}/b_0$; the mass matrix $\mathcal{M}$
stress tensor, radion, stabilizer, Friedmann equations	`feStress`, `uPot`	20	$\varepsilon = W/(a^3 b^4)$ …; $m_r^2 = \kappa K_U/12$; $8\pi G_4 = \kappa / V_\perp$
primordial regime, capture, decays, today, far future	`rIp1`, `rIq1`, `rIIr4`, `rIIIH`	21–23	the numbers of sections 7 and 11 of the document
natural couplings and their dilution	`dkLam`, `dkYeff`, `dkMuP`	24	$\lambda_{\rm eff} = c_P M_8/M_{\rm Pl}^2$, $y_{\rm eff} = c_y M_8/M_{\rm Pl}$
the auxiliary torus and the world brane	`tb…`	25	induced metrics, Christoffel symbols, U, brane spinors, smeared sources
the radion, the sphere, warped branes, Higgs–radion mixing	`rd…`	26	sections 7.5, 12 and 13 of the document

Before section 25 was appended, the evaluated notebook was copied to the first file below, before section 26 to the second; both are excluded from the repository by .gitignore (size and SHA-256):

```
backups/7+1_2026-10-28_before_torus_brane.nb
    1940664 bytes, SHA-256 39f64ee81d35c0c9fb94188fd7f8c8e75c96ea198fbf72a5ca6b3c0027426018
backups/7+1_2026-10-28_before_radion.nb
    2031601 bytes, SHA-256 44385a6b1d808f1f95b1d634d29c3960c6251cab96f436b9927522abe0fd8397
```

4. Data and their sources

quantity	value used	source
$\hbar c$	$1.973269804 \times 10^{-16}$ GeV m	CODATA 2018 (E. Tiesinga et al., Rev. Mod. Phys. 93 (2021) 025010)
G_F, m_W, m_Z, m_H	1.1663788×10^{-5} GeV^{-2}, 80.3692, 91.1880, 125.20 GeV	Particle Data Group 2025 (section 17.3)
H_0, Ω_m, Ω_r	67.4 km/s/Mpc, 0.315, 9.14×10^{-5}	Planck 2018 (Astron. Astrophys. 641 (2020) A6)
$M_{\rm Pl}$ (reduced), M_8, m_r	2.435×10^{18} GeV, 10 TeV, 10 GeV	as in sections 17–24
RS1 conventions	$2M^3 \int \sqrt{-G}R$, $V_{\rm hid} = -V_{\rm vis} = 24M^3k$	L. Randall, R. Sundrum, Phys. Rev. Lett. 83 (1999) 3370
Higgs–radion conventions	ξ, $Z^2 = 1 + 6\xi\gamma^2(1-6\xi)$	G. F. Giudice, R. Rattazzi, J. D. Wells, Nucl. Phys. B 595 (2001) 250
$\langle\mu\rangle$ of the warped example	3 TeV	the author's note

Software: Wolfram Language 15.0.1 (front end WolframNB.exe; the kernel configuration "Eval71" is a fresh kernel); Python 3.13.17 with numpy 2.5.3, scipy 1.18.1, sympy 1.14.0, mpmath 1.3.0.

5. Derivations

Induced metric. With $X^A(x) = (x_0, \dots, x_3, 0, 0, 0, 0)$, $J^A{}_\mu = \partial X^A/\partial x^\mu$ is the 8×4 matrix $(1_4; 0)$, and $h_{\mu\nu} = g_{AB}J^A{}_\mu J^B{}_\nu = g_{\mu\nu}(x, 0)$. For the cosmological metric $h = \mathrm{diag}(1, -a^2, -a^2, -a^2)$.

Extrinsic curvature. For $n_{(k)} = b^{-1}\partial_k$ and $g_{\mu k} = 0$: $K^{(k)}_{\mu\nu} = \frac{1}{2}(\mathcal{L}_{n_{(k)}}g)_{\mu\nu} = \frac{1}{2b}\partial_k g_{\mu\nu}$; and $\{k\,\mu\nu\} = -\frac{1}{2}g^{kk}\partial_k g_{\mu\nu} = \frac{1}{2b^2}\partial_k g_{\mu\nu}$ ($g^{kk} = -b^{-2}$). For the inhomogeneous metric at $x_0 = \frac{1}{2}$, $x_4 = 0$: $n = \frac{45}{44}$, $\partial_4 n = \frac{1}{10}$, $b = \frac{73}{36}$, so $\{4\,00\} = n\partial_4 n/b^2 = 1458/58619$.

Tension. $\delta\sqrt{|h|} = -\frac{1}{2}\sqrt{|h|}h_{\mu\nu}\delta h^{\mu\nu}$ gives $T^W_{\mu\nu} = \frac{2}{\sqrt{|h|}}\frac{\delta(-f_W\sqrt{|h|})}{\delta h^{\mu\nu}} = f_W h_{\mu\nu}$, hence $\varepsilon = f_W$, $p = -f_W$, $w = -1$.

Smeared sources. The first law $dW = -p_a\, d(a^3b^4)|_b - p_b\, d(a^3b^4)|_a$ gives $\varepsilon = W/(a^3b^4)$, $p_a = -\partial_a W/(3a^2b^4)$, $p_b = -\partial_b W/(4a^3b^3)$; a brane cell has the volume a^3, so W does not depend on b and $p_b = 0$.

Bulk radiation. For $W = \sqrt{k_a^2/a^2 + k_b^2/b^2}$ with momenta isotropic in seven dimensions, the angle between the three- and four-dimensional parts has the weight $\cos^2\theta \sin^3\theta$:

$$\langle\cos^2\theta\rangle = \frac{\int_0^{\pi/2}\cos^4\theta\sin^3\theta\,d\theta}{\int_0^{\pi/2}\cos^2\theta\sin^3\theta\,d\theta} = \frac{2/35}{2/15} = \frac{3}{7}, \qquad p_a = \frac{1}{3}\cdot\frac{3}{7}\varepsilon = \frac{\varepsilon}{7}, \qquad p_b = \frac{1}{4}\cdot\frac{4}{7}\varepsilon = \frac{\varepsilon}{7}.$$

Images and Kaluza–Klein modes. Poisson summation: $\sum_{\vec n}(r^2 + |\vec n|^2)^{-5/2} = \frac{4\pi^2}{3r}\sum_{\vec m} e^{-2\pi r|\vec m|}$ (lengths in units of $\ell_\perp$), from the four-dimensional Fourier transform $\int d^4y\, e^{-i\vec p\cdot\vec y}(r^2 + |\vec y|^2)^{-5/2} = \frac{4\pi^2}{3r}e^{-pr}$; with $8\pi G_4 = \kappa/V_\perp$ this gives $\Phi_{\rm N} \to -\frac{5}{3}G_4 M/r$.

Radion coupling. $h = e^{-4\beta}g_E$; a mass at rest: $-M\int e^{-2\beta}dx_0$, linear term $2M\beta = M\phi/(\sqrt{3}M_{\rm Pl})$; the potential between two masses $-\frac{MM'}{12\pi M_{\rm Pl}^2}\frac{e^{-m_r r}}{r} = -\frac{2}{3}\frac{G_4 MM'}{r}e^{-m_r r}$.

Brane spinors. $\Gamma_A U = U R_A$ with $R_\mu = \gamma_\mu \otimes 1_4$, $R_k = \gamma_5 \otimes \rho_k$; then $\Gamma_\mu\Gamma_\nu = U(\gamma_\mu\gamma_\nu \otimes 1_4)U^\dagger$, $\Gamma_k\Gamma_l = U(1_4 \otimes \rho_k\rho_l)U^\dagger$ ($\gamma_5^2 = 1_4$), and $\chi = i\Gamma_0\cdots\Gamma_7 = U(\gamma_5 \otimes \rho_{4567})U^\dagger$ from $R_0R_1R_2R_3 = -i\gamma_5 \otimes 1_4$ and $R_4R_5R_6R_7 = 1_4 \otimes \rho_{4567}$.

Kaluza–Klein masses at the brane. $(\gamma\cdot p\otimes 1_4+X)^2=(\gamma\cdot p)^2\otimes 1_4+X^2$ because γ_5 anticommutes with every γ_μ, and $X^2=-|\vec{q}|^2/b_0^2$ from $\rho_k\rho_l+\rho_l\rho_k=-2\delta_{kl}$: $m_{\vec{n}}^2=m^2+4\pi^2|\vec{n}|^2/\ell_\perp^2$.

Dilution of the portals. A brane coupling of a bulk field at $\vec{y}=0$ involves $\phi_{\vec{n}}(0)=1/\sqrt{V_\perp}$ per bulk field: $\lambda_{\rm eff}=\lambda_P/V_\perp$, $y_{\rm eff}=y/\sqrt{V_\perp}$, $G_{\rm eff}=G_P/V_\perp$; with $V_\perp=M_{\rm Pl}^2/M_8^6$ and natural couplings these are $c_P M_8/M_{\rm Pl}^2$, $c_y M_8/M_{\rm Pl}$, $c_V/M_{\rm Pl}^2$.

The normalcy temperature (ESTIMATE): the emission rate T^7/M_8^6 equals the expansion rate $T^2/M_{\rm Pl}$ at $T_*=(M_8^6/M_{\rm Pl})^{1/5}$.

The sphere brane. At the north pole of S^4 the induced metric of $ds^2=e^{-4\beta}g_E-e^{2\beta}r_0^2\hat{\gamma}$ is $e^{-4\beta}g_E$, as on the torus. The rotations $L_{ab}=X_a\partial_b-X_b\partial_a$ vanish at $(0,0,0,0,r)$ iff $a,b\le 4$ (6 of 10). The zonal harmonics are normalized by $\int_{S^4}(C_l/C_l(1))^2=V/d_l$ (exact for $l\le 6$), so $Z_l(0)=\sqrt{d_l/V}$. The Killing spinors satisfy $\nabla_i\varepsilon_s=s\frac{i}{2r}e_i{}^a\tilde{\gamma}_a\varepsilon_s$ exactly (checked after writing the angles as twice half-angles), hence $\not{D}\varepsilon_s=s\frac{2i}{r}\varepsilon_s$.

Warped branes. For $ds^2=e^{-2A}\eta-dy^2$: $G^\mu{}_\nu=3A''-6A'^2$; $A=k|y|$ gives $A''=\pm 2k\delta$ at the branes and the tensions $\pm 6k/\kappa$; with $K^\mu{}_\nu=\frac{1}{2}g^{\mu\alpha}\partial_y g_{\alpha\nu}=-A'\delta^\mu_\nu$, $[K^\mu{}_\nu]-\delta^\mu_\nu[K]=3[A']$ is the δ part of $G^\mu{}_\nu$. In eight dimensions with warping along one direction, $G^z{}_z=4A''-10A'^2\neq G^y{}_y=-6A'^2$ unless $A'=0$.

Higgs–radion. For $h=\Omega^2\eta$, $\sqrt{|h|}R(h)=-6\Omega\Box\Omega$ (computed); with $\Omega=1+o_1\phi+\dots$ and $\varphi_H=v+h$ the curvature term has the quadratic part $-3\xi(v^2o_1^2\phi\Box\phi+2vo_1h\Box\phi)$, which after integrations by parts is $3\xi\gamma^2(\partial\phi)^2+6\xi\gamma\,\partial h\cdot\partial\phi$ ($\gamma=vo_1$); the shift and the rotation then give Z^2 and $\tan 2\theta$ of the document.

6. Verification

6.1 The checks of sections 25 and 26 of 7+1.nb

Evaluation of 7+1.nb: Wolfram 15.0.1 for Microsoft Windows (64-bit) (July 2, 2026), a fresh kernel (front-end kernel configuration Eval71); finished Wed 28 Oct 2026 08:57:47; 283.883 s; message cells 0; output cells 554. The checks of section 25 (1–14 the torus, 15–16 torus and brane, 17–27 the world brane):

#	check	result	section
1	auxiliary torus: for the cosmological metric det(-g_kl) == b^8 (k, l = 4..7), and the proper volume of the fundamental domain [0, Lc)^4, Integrate[Sqrt[det(-g_kl)], {y, 0, Lc}^4], is b^4 Lc^4 = V_perp	True	25.1
2	only the product b Lc of the auxiliary torus is physical: y = lambda y' maps the torus of period Lc and size b to the one of period Lc/lambda and size lambda b; the line element b^2 dy.dy, the proper period b Lc and V_perp = b^4 Lc^4 are unchanged	True	25.1
3	symmetries of the auxiliary torus that fix y = 0 (and the brane): the 384 = 2^4 4! signed permutation matrices are distinct orthogonal integer matrices (they map Z^4 onto itself and preserve |n|^2); the shells |n|^2 = 1, 2, 3, 4, 9 split into orbits of sizes {8}, {24}, {32}, {8, 16}, {8, 96}	True	25.1
4	Fourier modes of the auxiliary torus: phi_n = Exp[I q_n.y]/Sqrt[V_perp] with q_n = 2 Pi n/Lc (n in Z^4) are orthonormal for the proper volume element b^4 d^4y (exact integrals: n = m = (1, 0, -2, 3); n = (1, 0, -2, 3), m = (1, 1, -2, 3); n = 0, m = (0, 2, 0, 0))	True	25.2
5	Parseval on the auxiliary torus: f(y) = Prod_k cos^2(Pi y_k/Lc) = Prod_k (1/2 + (1/4) e^(2 Pi I y_k/Lc) + (1/4) e^(-2 Pi I y_k/Lc)) has the 81 non-zero coefficients c_n = Sqrt[V_perp] Prod_k w(n_k), w(0) = 1/2, w(+-1) = 1/4 (three of them checked by integration), and Sum_n |c_n|^2 == Int b^4 d^4y f^2 == (3/8)^4 V_perp	True	25.2
6	the Laplacian of the auxiliary torus: -(1/b^2) Sum_k d^2 phi_n/dy_k^2 == (|q_n|^2/b^2) phi_n (n = (2, -1, 0, 3))	True	25.2
7	scalar Kaluza-Klein masses: for the static metric diag(1, -1, -1, -1, -b0^2, ..., -b0^2) the Klein-Gordon operator Box + m^2 (Box f = |g|^(-1/2) d_A(|g|^(1/2) g^AB d_B f)) on Exp[-I E x0 + I k.x + I q.y] gives (-E^2 + |k|^2 + |q|^2/b0^2 + m^2) times the wave: in ordinary spacetime a particle of mass Sqrt[m^2 + |q|^2/b0^2]	True	25.2
8	Jacobi's four-square theorem: the number of n in Z^4 with |n|^2 = k is r_4(k) = 8 sigma(k) - 32 sigma(k/4) for k = 1..100 (direct count over {-10, ..., 10}^4)	True	25.3
9	Weyl's law: the number N(R) of n in Z^4 with |n| <= R approaches the volume Pi^2 R^4/2 of the four-ball: N(10) = 49689, N(100) = 493490641, and |N(R)/(Pi^2 R^4/2) - 1| < 10^-2, 10^-3, 3 10^-5, 5 10^-6, 10^-6 for R = 10, 30, 100, 300, 1000	True	25.3
10	the auxiliary torus today (M8 = 10 TeV, M_Pl = 2.435 10^18 GeV): V_perp = M_Pl^2/M8^6 = 5.929 10^12 GeV^-4, l_perp = V_perp^(1/4) = b0 Lc = 1560.4 GeV^-1 = 3.079 10^-13 m, lowest Kaluza-Klein mass 2 Pi/l_perp = 4.027 MeV; N(mu) = V_perp mu^4/(32 Pi^2) = 1.17 10^13 levels below m_r/2 = 5 GeV (the lattice count of section 22.2 agrees to 10^-5) and N(M8) = M_Pl^2/(32 Pi^2 M8^2)	True	25.3

#	check	result	section
11	Newton's law in eight dimensions: Omega_6 = 2 Pi^(7/2)/Gamma(7/2) = 16 Pi^3/15, r^-5 is harmonic away from r = 0 in seven space dimensions ((1/r^6) d/dr (r^6 d/dr r^-5) = 0), and grad^2 Phi = 8 Pi G8 ((D - 3)/(D - 2)) M delta^7 (D = 8) gives Phi = -(5/6) kappa M/(5 Omega_6 r^5) = -5 kappa M/(32 Pi^3 r^5); the same formula in D = 4 gives -G M/r	True	25.4
12	images = Kaluza-Klein modes (Poisson summation; lengths in units of l_perp): Sum_n (r^2 + |n|^2)^(-5/2) == (4 Pi^2/(3 r)) Sum_m Exp[-2 Pi r |m|] to 10^-8 at r = 1/4, 1/2, 1, 2 (left: shells with r_4(k) to k = 1.5 10^6 and the continuum tail Pi^2 Int k (r^2 + k)^(-5/2) dk; right: shells to k = 20000)	True	25.4
13	near and far field: Z(5/2) = Sum_(n != 0) |n|^-5 = Sum_k r_4(k) k^(-5/2) = 7 zeta(5/2) zeta(3/2) = 24.531 (shell sum with tail, to 10^-8); near the source r^5 S(r) - 1 -> r^5 Z(5/2) (the ratio lies between 0.98 and 1 at r = 1/10 and 1/20, closer to 1 at 1/20): the eight-dimensional law; far from it 3 r S/(4 Pi^2) - 1 -> 8 Exp[-2 Pi r], the 8 lightest Kaluza-Klein modes (ratio 1 to 2 10^-2 at r = 2 and 3)	True	25.4
14	graviton + radion: far from the source the images add up to (5 kappa M/(32 Pi^3)) (4 Pi^2/(3 V_perp r)) == (5/3) G4 M/r with G4 = kappa/(8 Pi V_perp); the radion of section 20.3, coupled to matter by (phi/(Sqrt[3] M_Pl)) T, gives the potential -(1/(Sqrt[3] M_Pl))^2 M M' Exp[-m_r r]/(4 Pi r), and (1/(Sqrt[3] M_Pl))^2/(4 Pi) == (2/3) G4 (8 Pi G4 = 1/M_Pl^2): graviton 1 + massless radion 2/3 = 5/3	True	25.4
15	Kaluza-Klein momentum is conserved in the bulk and not at the brane: Int b^4 d^4y phi_n phi_m phi_l^* == 1/Sqrt[V_perp] for l = n + m and 0 otherwise (exact: n = (1, 0, 0, 2), m = (0, 1, -1, 0), l = (1, 1, -1, 2) and (1, 1, -1, 1)); at y = 0, phi_n phi_m phi_l^* == V_perp^(-3/2) for every n, m, l	True	25.5
16	the delta function of the torus: Int_(-Lc/2)^(Lc/2) Exp[-I q_n y] delta(y) dy == 1 for every integer n, so delta^4(y) = Lc^-4 Sum_n Exp[I q_n.y] (every mode with the weight 1/Lc^4); a brane source rho4 delta^4(y)/b^4 has Int b^4 d^4y (rho4 delta^4(y)/b^4) == rho4, i.e. the average rho4/V_perp over the torus	True	25.5
17	the world brane W = {x4 = x5 = x6 = x7 = 0}: the induced metric h = J^T.g.J of the embedding X = (x0, x1, x2, x3, 0, 0, 0, 0) is, for the cosmological metric of section 11.2, the Friedmann-Robertson-Walker metric diag(1, -a^2, -a^2, -a^2) (independent of b); for the inhomogeneous metric of section 11.3 it is diag(n^2, -a^2, -a^2, -a^2) at x4 = 0 with n = 1 + x0^2/11, a = 1 + x0/3	True	25.6
18	brane integrals in eight-dimensional form: at the brane Sqrt|det g| == Sqrt|det h| b^4 with b^4 = Sqrt[det(-g_kl)] (cosmological metric; inhomogeneous metric exactly at x0 = 1/2), so Int_W d^4x Sqrt|h| L == Int d^8X Sqrt|g| (delta^4(y)/b^4) L; test: Int d^4y b^4 (delta^4(y)/b^4) F(y) == F(0) = 1 for F = Exp[x4 + 2 x5] Cos[x6 - x7]	True	25.6
19	how the brane sits in the bulk: the Christoffel symbols {k mu nu} at the brane (k = 4..7, mu, nu = 0..3; = (1/(2 b^2)) d_k g_mu nu for these metrics, b times the extrinsic curvature) all vanish for the cosmological metric: the brane is totally geodesic, and a free particle moving along it stays on it. For the inhomogeneous metric at x0 = 1/2 the only non-zero one is {4 00} = n (d_4 n)/b^2 = 1458/58619: a free particle at rest on the brane is accelerated off it	True	25.6
20	explicit forms: every Gamma17[A] is a signed permutation matrix (one entry +-1 in each row and column), and 2 Sqrt[2] U has only the entries 0, +-1, +-I, eight non-zero ones in each row and in each column	True	25.7
21	brane spinor (x) transverse spinor: Gamma17[mu].Gamma17[nu] == U.(gamma_mu.gamma_nu (x) 1_4).U^dag, Gamma17[k].Gamma17[l] == U.(1_4 (x) rho_k.rho_l).U^dag and Gamma17[mu].Gamma17[k] == U.(gamma_mu.gamma5 (x) rho_k).U^dag (mu, nu = 0..3, k, l = 4..7): the Lorentz group of the brane acts on the brane spinor index, the rotations about the brane on the transverse index, and the generators that mix them are broken by the brane	True	25.7
22	the transverse index carries Spin(4) = SU(2) x SU(2): rho4567 == -gamma5 (4 x 4), it commutes with every rho_k.rho_l and anticommutes with every rho_k; for each even permutation (k, l, m, n) of (4, 5, 6, 7) the combination rho_k.rho_l + s rho_m.rho_n vanishes on the eigenspace rho4567 = s, and rho_k.rho_l - s rho_m.rho_n acts there with rank 2: the four values of a form (2, 1) + (1, 2)	True	25.7
23	chirality at the brane: PL17 == U.(P_L (x) P_+ + P_R (x) P_-).U^dag with P_L,R = (1 -+ gamma5)/2 and P_+- = (1 +- rho4567)/2: the left-handed half psi_L (chi17 = -1, 8 components) is two brane-left-handed Weyl spinors with rho4567 = +1 and two brane-right-handed ones with rho4567 = -1 (each part of rank 4)	True	25.7
24	an explicit spinor: psi = U.(u (x) e_1) with the brane spinor u = (1, 2, 0, -1): psibar psi == ubar u == 4 and psibar Gamma^mu psi == ubar gamma^mu u (mu = 0..3)	True	25.7
25	brane tension: for S_W = -Int d^4x Sqrt|h| f_W and h = diag(1/u0, -1/u1, -1/u2, -1/u3) (h^00 = u0, h^ii = -u_i), T_mu nu = (2/Sqrt|h|) dS_W/dh^mu nu == f_W h_mu nu, T^0_0 == f_W (energy density) and T^1_1 == f_W (pressure -f_W: w = -1 along the brane)	True	25.8

#	check	result	section
26	sources smeared over the torus (W(a, b) the comoving energy, section 20.2): the brane tension W = a^3 f_W/Lc^4 gives eps = f_W/V_perp, pa = -eps, pb = 0; brane radiation W = a^3 r0 a^-4/Lc^4 gives pa = eps/3, pb = 0; radiation of the bulk, isotropic in the seven space directions, has <cos^2 theta> = 3/7 and <sin^2 theta> = 4/7 for the weight cos^2 theta sin^3 theta of the momentum sphere, so pa = (1/3)(3/7) eps = eps/7 and pb = (1/4)(4/7) eps = eps/7	True	25.8
27	the brane in four-dimensional physics: (i) the tension is vacuum energy, kappa f_W/V_perp == f_W/M_Pl^2 (V_perp = kappa M_Pl^2), and f_W = M8^4 is between 3 10^62 and 5 10^62 times rho_Lambda = 3 H0^2 M_Pl^2 Omega_Lambda (section 24.5); (ii) normalcy temperature: (T^7/M8^6)/(T^2/M_Pl) == 1 at T* = (M8^6/M_Pl)^(1/5) = 13.3 GeV, and V_perp T^4 T^3/M_Pl^2 == T^7/M8^6; (iii) thin brane: (1/y^3) d/dy (y^3 d/dy y^-2) == 0 and 2 Pi^2 y^3 d/dy y^-2 == -4 Pi^2 (the Green's function -1/(4 Pi^2 |y|^2) of the four transverse directions), and Sqrt[kappa f_W] == 1/M8 for f_W = M8^4	True	25.8

The checks of section 26 (the radion; sections 7.5, 12 and 13 of the document):

#	check	result	section
1	section 26 engine for diagonal metrics: rdScal and rdEin reproduce curvature[] of section 11 for the cosmological metric (R and every G^A_A)	True	26.1
2	the 4-sphere of radius r0: Ricci_ij = 3 gamma_ij (gamma the unit-sphere metric), Ricci_ij = 0 for i != j, so R(S^4) = 12/r0^2 (in the signature of 7+1.nb the space-like block gives R = -12/r0^2, also with curvature[] of section 11); the volume of the unit S^4 is 8 Pi^2/3	True	26.1
3	radion on T^4: for ds^2 = e^(-4 beta) (N^2 dx0^2 - a^2 dx.dx) - e^(2 beta) b0^2 dy.dy the reduced action -(1/(2 kappa)) Int(R + 2 Lambda) - Int U equals -(M_Pl^2/2) R_E + 6 M_Pl^2 (d beta)^2 - e^(-4 beta)(M_Pl^2 Lambda + V0 U) up to a total derivative (Euler-Lagrange expressions for N, a, beta vanish), M_Pl^2 = V0/kappa, V0 = b0^4 Lc^4	True	26.1
4	radion on S^4: the same reduction for ds^2 = e^(-4 beta) g_E - e^(2 beta) r0^2 gamma (unit-sphere gamma) holds with the additional potential -(M_Pl^2/2) e^(-4 beta) 12/(r0^2 e^(2 beta)) = -6 M_Pl^2 e^(-6 beta)/r0^2 from the curvature of the sphere, V0 = (8 Pi^2/3) r0^4; R is independent of the angles	True	26.1
5	n extra dimensions (torus T^n, D = 4 + n, n = 1..4, Einstein frame e^(-n B) g_E): the kinetic term is (M_Pl^2/2)(n(n+2)/2)(dB)^2; with the note's B = beta_n phi, beta_n = Sqrt[2/(n(n+2))], it is (M_Pl^2/2)(d phi)^2, not (M_Pl^2/2)(1/2)(d phi)^2; the form with 1/2 needs beta_n = 1/Sqrt[n(n+2)]	True	26.1
6	the note's eight-dimensional values alpha = -1/Sqrt[3], beta = 1/Sqrt[12]: 2 alpha + 4 beta == 0 (Einstein frame), 4 beta == 2/Sqrt[3] (V = V0 e^(4 beta phi)), and Sqrt[12] beta == 1, so M_Pl phi_note = Sqrt[12] M_Pl ln(b/b0) is the canonical radion of section 20.3	True	26.1
7	the cosmological metric with the sphere of radius b(x0) (ds^2 = dx0^2 - a^2 dx.dx - b^2 dOmega_4^2): G^A_A = (torus value) + (6, 6, 6, 6, 3, 3, 3, 3)/b^2 and R = (torus value) - 12/b^2	True	26.2
8	radion equation on S^4 (Einstein gravity; 11 and 44 equations with the 00 constraint): Hb' + 3 Ha Hb + 4 Hb^2 == kappa (eps/6 - pa/2 + pb/3) + Lambda/3 - 3/b^2	True	26.2
9	radion on T^4 with U = U0 + U1 beta + (K_U/2) beta^2: a flat vacuum at b = b0 needs Lambda + kappa U0 = 0 and U1 = 0, and m_r^2 = V''(beta)/(12 M_Pl^2) = kappa K_U/12 (section 20.3)	True	26.2
10	radion on S^4 with U: Lambda + kappa U0 == 6/r0^2, kappa U1 == -12/r0^2 and m_r^2 == kappa K_U/12 - 2/r0^2 from V(beta); the eight-dimensional radion equation gives the same static solution and the same mass	True	26.2
11	S^4 with Lambda only: dS4 x S^4 with r0^2 = 9/Lambda, H^2 = 1/r0^2 = Lambda/9 solves the 00, 11 and 44 equations; it is a MAXIMUM of V: m^2 = -6 H^2 from V(beta), and m^2 = -6/r0^2 = -6 H^2 from the eight-dimensional radion equation	True	26.2
12	S^4 with a 4-form flux (illustration; W = a^3 b^4 q^2/(2 b^8) gives eps = rho_F, pa = -rho_F, pb = +rho_F): Minkowski x S^4 needs Lambda = 9/(2 r0^2) and kappa rho_F = 3/(2 r0^2); m^2 = 6/r0^2 from V(beta) and from the eight-dimensional radion equation	True	26.2
13	the note's Goldberger-Wise potential in eight dimensions: for a static profile Phi(y) (here Phi(x4, x5)) Sqrt|G| (1/2 G^AB dPhi dPhi - 1/2 m^2 Phi^2) == -N a^3 e^(-4 beta) Sqrt[gamma] (1/2 e^(-2 beta) |grad Phi|^2_gamma + 1/2 m^2 Phi^2): the Einstein-frame factor is e^(-4 beta) (e^(-4 beta phi) in the note's notation), not e^(+4 beta phi)	True	26.2
14	the radion couples to a mass M on the brane as (phi/(Sqrt[3] M_Pl)) M: -M e^(-n beta/2) = -M + (n/2) M beta + ..., with beta = phi/(M_Pl Sqrt[n (n + 2)/2]) the coupling is Sqrt[n/(2 (n + 2))]/M_Pl = 1/(Sqrt[3] M_Pl) for n = 4, and the fifth-force strength 2 (coupling M_Pl)^2 = n/(n + 2) = 2/3	True	26.3

#	check	result	section
15	Eot-Wash (|alpha| = 1 excluded for lambda > 38.6 micrometres): a radion of strength 2/3 must be heavier than about hbar c/38.6 micrometres = 5.1 meV; the 10 GeV radion has the range 2.0 10^-17 m; Higgs-radion mixing in 7+1 is governed by gamma = v/(Sqrt[3] M_Pl) = 5.8 10^-17	True	26.3
16	zonal harmonics on S^4 (l = 0..6, exact): (1/sin^3 x4) d/dx4 (sin^3 x4 d/dx4 C_l^(3/2)(cos x4)) == -l (l + 3) C_l^(3/2), C_l^(3/2)(1) == (l + 1)(l + 2)/2, and Int over S^4 of (C_l/C_l(1))^2 == V/d_l (unit sphere), so the normalized zonal harmonic is Sqrt[d_l/V] at the north pole; d_l == Binomial[l + 4, 4] - Binomial[l + 2, 4] for l = 0..20	True	26.4
17	Weyl's law on S^4: the number of levels with l(l + 3) <= (mu r)^2 approaches (mu r)^4/12 = V mu^4/(32 Pi^2) (V = 8 Pi^2 r^4/3): relative deviation -1.0 10^-2, -1.0 10^-4, -1.0 10^-6 at mu r = 10, 100, 1000 (like -1/(mu r)^2)	True	26.4
18	the brane at the north pole of S^4: the embedding (r0 sin x4 sin x5 sin x6 cos x7, ..., r0 cos x4) of S^4 in R^5 induces r0^2 gamma; the north pole x4 = 0 is (0, 0, 0, 0, r0); of the 10 rotations L_ab of R^5 (the Killing vectors of S^4, SO(5)) the 6 with a, b <= 4 vanish there (SO(4), kept by the brane) and the 4 L_a5 move it	True	26.4
19	Killing spinors of S^4 (radius r0, frame e^a = (r0, r0 sin x4, ...) dx^(a+3), gt_a = I rho_(a+3)): eps_s = exp(s (i/2) x4 gt_1) exp((1/2) x5 gt_1 gt_2) exp((1/2) x6 gt_2 gt_3) exp((1/2) x7 gt_3 gt_4) eps0 satisfies nabla_mu eps_s == s (i/(2 r0)) e_mu^a gt_a eps_s (s = +-1, all eps0), so the Dirac operator gives (2 s i/r0) eps_s: |lambda| = 2/r0, Friedrich's bound n R/(4(n - 1)) = 4/r0^2	True	26.4
20	Dirac spectrum on S^4: Weyl's law for spinors (4 components) N(|lambda| <= L/r) ~ 4 V (L/r)^4/(32 Pi^2) = L^4/3 agrees with the degeneracies 2 x 4 Binomial[l + 3, 3] of the eigenvalues +-i(l + 2)/r (to 2.0 10^-3 at L = 1000)	True	26.4
21	the sphere with the volume of the torus today (V = M_Pl^2/M8^6): r = (3 V/(8 Pi^2))^(1/4) = 688.9 GeV^-1 = 1.359 10^-13 m; lowest scalar levels 4.03 MeV (torus, 8 states) and 2.90 MeV (sphere, l = 1, 5 states); lowest dirac16complex level on the sphere 2/r = 2.90 MeV (8 Dirac fermions; the torus has 4 massless ones for m = 0); levels below 5 GeV: torus 1.17335 10^13, sphere 1.17375 10^13, Weyl 1.17335 10^13	True	26.4
22	RS1 in five dimensions (mostly minus): G^mu_nu = 3 A'' - 6 A'^2, G^y_y = -6 A'^2; bulk A = k y needs Lambda = -6 k^2; the delta functions A'' = +-2 k delta at the branes need the tensions f_UV = +6 k/kappa, f_IR = -6 k/kappa (UV positive)	True	26.5
23	Israel junction at the UV brane of RS1: with K^mu_nu = (1/2) g^mu alpha d_y g_alpha nu = -A' delta^mu_nu (normal d_y), [K^mu_nu] - delta^mu_nu [K] = 6 k = (coefficient of A'' in G^mu_nu) x (jump of A'), the delta-function part of G^mu_nu; in the signature (-,+,+,+,+) of the note G^A_B changes sign, so there [K_mu nu - g_mu nu K] = -kappa S_mu nu (the note uses +kappa S_mu nu, hence its reversed tensions)	True	26.5
24	RS1: M_Pl^2 = (1/kappa) Int_(-Pi R)^(Pi R) e^(-2 k |y|) dy = (1 - e^(-2 Pi k R))/(kappa k)	True	26.5
25	RS1 radion: with R promoted to T(x), the bulk action is (3/(kappa k)) (d e^(-k Pi T))^2 + V(T) + (a total derivative), so the canonical radion is mu = Sqrt[6/(kappa k)] e^(-k Pi T) (the note: Sqrt[12/(kappa k)])	True	26.5
26	eight dimensions, warped along one extra direction (ds^2 = e^(-2 A(y)) eta - dy^2 - dz.dz): G^mu_nu = 3 A'' - 6 A'^2, G^y_y = -6 A'^2, G^z_z = 4 A'' - 10 A'^2; with only Lambda (all G^A_A equal) A'' = 0 and A' = 0: the note's five-dimensional equations do not carry over	True	26.5
27	Goldberger-Wise in RS1: the static bulk scalar (Sqrt|G| = e^(-4 k y), G^yy = -1) obeys Phi'' - 4 k Phi' - m^2 Phi = 0, solved by e^(2 k y)(C1 e^(nu k y) + C2 e^(-nu k y)), nu = Sqrt[4 + m^2/k^2] = 2 + eps + O(m^4), eps = m^2/(4 k^2)	True	26.5
28	Goldberger-Wise radion mass: V(mu) = 4 k (mu/f)^4 (v_B - v_A (mu/f)^eps)^2 has V'(<mu>) = 0 at (<mu>/f)^eps = v_B/v_A (k Pi <T> = (4 k^2/m^2) Log[v_A/v_B]) and V''(<mu>) = 8 k eps^2 v_B^2 <mu>^2/f^4; with f^2 = 6/(kappa k): m_r^2 = (4/3) kappa k^2 eps^2 v_B^2 (<mu>/f)^2 (the note's f^2 = 12/(kappa k) gives 2/3)	True	26.5
29	conformally flat brane metric h = Omega^2 eta (four dimensions, Omega(x0, x1)): Sqrt|h| R(h) == -6 Omega box Omega with box = d0^2 - d1^2 (curvature conventions of 7+1.nb)	True	26.6
30	the Higgs boson on the brane (h = Omega^2 eta, Omega = 1 + r/Lambda + a2 r^2, phi_H = v + h, gamma = v/Lambda): the terms linear in (h, r) are total derivatives, and to second order L == (1/2)(dh)^2 + (1/2)(1 + 6 xi gamma^2)(dr)^2 + 6 xi gamma dh.dr - (1/2) m_h0^2 h^2 - (1/2) m_r0^2 r^2 up to total derivatives (Euler-Lagrange expressions in x0 and x1); a2 drops out	True	26.6
31	canonical kinetic terms: Transpose[S].K.S == 1 for S = {{1, -6 xi gamma/Z}, {0, 1/Z}} if and only if Z^2 = 1 + 6 xi gamma^2 (1 - 6 xi) (the note's Z); with the note's literal kinetic terms (no 6 xi gamma^2 for the radion) it would be Z^2 = 1 - 36 xi^2 gamma^2	True	26.6

#	check	result	section
32	mass mixing: with (h', r') = Q(theta).(h_m, r_m), Q = {{cos, -sin}, {sin, cos}}, the off-diagonal element of Transpose[Q].M'.Q is -(1/2)(M'_11 - M'_22) sin 2 theta + M'_12 cos 2 theta, zero for tan 2 theta = 2 M'_12/(M'_11 - M'_22) = 12 gamma xi Z m_h0^2/(m_r0^2 - m_h0^2 (Z^2 - 36 xi^2 gamma^2)) (the note's formula); with the inverse matrix (the note's display) it is +(1/2)(M'_11 - M'_22) sin 2 theta + M'_12 cos 2 theta (theta -> -theta); (h, r) = S.Q.(h_m, r_m) gives C_h = cos theta - (6 xi gamma/Z) sin theta, C_r = sin theta/Z	True	26.6
33	Higgs-radion mixing in 7+1 (gamma = v/(Sqrt[3] M_Pl), xi = 1, m_r0 = 10 GeV, m_h0 = 125.20 GeV; 60-digit arithmetic): |Z - 1| < 10^-30, |theta| = 3.5 10^-16, the rates of h -> gamma gamma and h -> g g change by less than 10^-30; in RS1 with <mu> = 3 TeV gamma = v/(3 TeV) = 0.082	True	26.6

6.2 The independent Python programs

tools/ref_torus_brane.py — its boolean checks:

#	check	result
1	torus: det(-g_kl) = b^8 for the cosmological metric; the proper volume of [0, Lc)^4 is b^4 Lc^4	True
2	only b Lc is physical: y = lambda y' maps (Lc, b) to (Lc/lambda, lambda b); b^2 dy.dy, b Lc and b^4 Lc^4 unchanged	True
3	symmetries fixing y = 0: 384 = 2^4 4! distinct signed permutation matrices, orthogonal integer matrices; shells 1, 2, 3, 4, 9 have orbits {8}, {24}, {32}, {8, 16}, {8, 96}	True
4	Fourier modes phi_n = exp(i q_n.y)/sqrt(V_perp), q_n = 2 pi n/Lc: orthonormal for b^4 d^4y (three exact pairs)	True
5	Parseval: f = prod cos^2(pi y_k/Lc) has 81 coefficients sqrt(V_perp) prod w(n_k) (3 checked by integration), Sum |c_n|^2 = Int b^4 f^2 = (3/8)^4 V_perp	True
6	Laplacian of the torus: -(1/b^2) Sum d^2/dy_k^2 phi_n = (|q_n|^2/b^2) phi_n	True
7	scalar Kaluza-Klein masses: (Box + m^2) exp(-iEx0 + ik.x + iq.y) = (-E^2 + |k|^2 + |q|^2/b0^2 + m^2) exp(...) on diag(1,-1,-1,-1,-b0^2,...)	True
8	graviton on the torus: D(D-3)/2 = 20 (D = 8) = 2 + 8 + 10 (zero mode: graviton, 4 graviphotons, 10 moduli) = 5 + 9 + 6 (massive level)	True
9	Weyl's law: N(10) = 49689, N(100) = 493490641; |N(R)/(pi^2 R^4/2) - 1| < 1e-2, 1e-3, 3e-5, 5e-6, 1e-6	True
10	numbers today: V_perp = 5.929e12 GeV^-4, l_perp = 1560.4 GeV^-1 = 3.079e-13 m, 2 pi/l_perp = 4.027 MeV; N(5 GeV) = 1.17e13 (lattice to 1e-5); N(M8) = M_Pl^2/(32 pi^2 M8^2)	True
11	from eight to four dimensions: for diag(1, -a^2, -a^2, -a^2, -b0^2, ..., -b0^2) (static torus) R = R4 = -6(Ha' + 2 Ha^2), the scalar curvature of the FRW metric diag(1, -a^2, -a^2, -a^2); sqrt|g| = a^3 b0^4	True
12	quaternion subalgebra of the brane directions: e1 e2 = -e3, e2 e3 = -e1, e3 e1 = -e2 (e_i e_j = -delta_ij - eps_ijk e_k), span{e0, ..., e3} closed under the octonion product, Re(e_A e_B) = diag(1, -1, -1, -1)	True
13	Newton in 8D: Omega_6 = 16 pi^3/15, (1/r^6) d/dr (r^6 d/dr r^-5) = 0, Phi = -(5/6) kappa M/(5 Omega_6 r^5) = -5 kappa M/(32 pi^3 r^5); D = 4: -G M/r	True
14	4D Fourier transform: Int d^4y exp(-ip.y) (r^2 + |y|^2)^(-5/2) = (4 pi^2/(3 r)) exp(-p r) (mpmath, three (p, r), to 1e-20)	True
15	Poisson duality: Sum_n (r^2 + |n|^2)^(-5/2) = (4 pi^2/(3 r)) Sum_m exp(-2 pi r |m|) to 1e-8 at r = 1/4, 1/2, 1, 2	True
16	Z(5/2) = 7 zeta(5/2) zeta(3/2) = Sum r_4(k) k^-5/2 (to 1e-8); near field ratio -> 1 (0.98 < . < 1 at r = 1/10, 1/20); far field ratio -> 1 (to 2e-2 at r = 2, 3)	True
17	graviton + radion: (5 kappa M/(32 pi^3))(4 pi^2/(3 V_perp r)) = (5/3) G4 M/r; radion (1/(sqrt3 M_Pl))^2/(4 pi) = (2/3)/(8 pi M_Pl^2)	True
18	Kaluza-Klein momentum: bulk vertex Int b^4 phi_n phi_m phi_l^* = 1/sqrt(V_perp) for l = n + m, 0 otherwise; at y = 0 phi_n phi_m phi_l^* = V_perp^(-3/2)	True
19	delta function of the torus: Fourier coefficients Int exp(-i q_n y) delta(y) dy = 1; Int b^4 d^4y rho4 delta^4(y)/b^4 = rho4 (average rho4/V_perp)	True
20	induced metric h = J^T g J: FRW diag(1, -a^2, -a^2, -a^2) for the cosmological metric; diag(n^2, -a^2, -a^2, -a^2) at x4 = 0 with n = 1 + x0^2/11, a = 1 + x0/3 for the inhomogeneous metric	True
21	brane volume element: sqrt|g| = sqrt|h| b^4 at the brane (cosmological metric; inhomogeneous metric at x0 = 1/2, exact), so Int_W d^4x sqrt|h| L = Int d^8X sqrt|g| delta^4(y)/b^4 L; test Int b^4 (delta^4/b^4) F = F(0)	True
22	Christoffel symbols {k mu nu} at the brane: all 0 for the cosmological metric (totally geodesic); for the inhomogeneous metric at x0 = 1/2 only {4 00} = 1458/58619 = n d4n/b^2	True

#	check	result
23	R_A = (gamma_mu (x) 1_4, gamma5 (x) rho_k) and Gamma17[A] both satisfy X_A X_B + X_B X_A = 2 eta_AB 1_16	True
24	every Gamma17[A] is a signed permutation matrix	True
25	intertwiner: Schur average with the seed matrix unit (1, 2); 2 sqrt(2) U has entries 0, +-1, +-i (rounding error < 1e-12) with 8 non-zero entries in each row and column; Gamma17[A].U8 = U8.R_A and U8^dag.U8 = 8 1_16 exactly	True
26	Schur's lemma: the linear map X -> (Gamma_A X - X R_A)_A has a one-dimensional null space (one singular value < 1e-10, the next > 0.1)	True
27	brane Lorentz group on the brane spinor index, transverse rotations on a: Gamma_mu Gamma_nu = U (gamma_mu gamma_nu (x) 1) U^dag, Gamma_k Gamma_l = U (1 (x) rho_k rho_l) U^dag, Gamma_mu Gamma_k = U (gamma_mu gamma5 (x) rho_k) U^dag	True
28	Spin(4) on the transverse index: rho4567 = -gamma5 (4x4) commutes with rho_k rho_l, anticommutes with rho_k; (rho_k rho_l + s rho_m rho_n) P_s = 0 and (rho_k rho_l - s rho_m rho_n) P_s has rank 2 for even (k,l,m,n): 4 = (2,1) + (1,2)	True
29	chirality: chi = i Gamma_0...Gamma_7 = U (gamma5 (x) rho4567) U^dag; P_L17 = U (P_L (x) P_+ + P_R (x) P_-) U^dag, each part of rank 4	True
30	example psi = U.(u (x) e_1), u = (1, 2, 0, -1): psibar psi = ubar u = 4 and psibar Gamma^mu psi = ubar gamma^mu u	True
31	four brane Dirac spinors: psibar psi' = Sum_a psibar_a psi'_a and psibar Gamma^mu psi' = Sum_a psibar_a gamma^mu psi'_a (random integer spinors)	True
32	Kaluza-Klein operator X = gamma5 (x) (rho.q)/b0: X^2 = -|q|^2/b0^2 1_16 and X anticommutes with gamma_mu (x) 1_4	True
33	brane tension: T_mu nu = (2/sqrt|h|) dS/dh^mu nu = f_W h_mu nu for S = -Int sqrt|h| f_W; T^0_0 = f_W, T^1_1 = f_W (p = -f_W, w = -1)	True
34	smeared sources: tension eps = f_W/V_perp, pa = -eps, pb = 0; brane radiation pa = eps/3, pb = 0; bulk radiation <cos^2> = 3/7, <sin^2> = 4/7: pa = pb = eps/7	True
35	normalcy temperature: (T^7/M8^6)/(T^2/M_Pl) = 1 at T* = (M8^6/M_Pl)^(1/5) = 13.3 GeV; V_perp T^4 T^3/M_Pl^2 = T^7/M8^6	True
36	the tension is vacuum energy: kappa f_W/V_perp = f_W/M_Pl^2 (V_perp = kappa M_Pl^2); f_W = M8^4 is 3e62 to 5e62 rho_Lambda	True
37	Green's function of the four transverse directions: (1/y^3) d/dy (y^3 d/dy y^-2) = 0, flux 2 pi^2 y^3 d/dy y^-2 = -4 pi^2; sqrt(kappa f_W) = 1/M8 for f_W = M8^4	True

tools/ref_radion.py — its boolean checks:

#	check	result
1	the 4-sphere of radius r0: Ricci_ij = 3 gamma_ij, off-diagonal 0, R = -12/r0^2 in the signature of 7+1.nb; volume of the unit S^4 = 8 pi^2/3	True
2	radion on T^4: the reduced action equals -(M_Pl^2/2) R_E + 6 M_Pl^2 (d beta)^2 - e^(-4 beta)(M_Pl^2 Lambda + V0 U) up to a total derivative (Euler-Lagrange for N, a, beta), M_Pl^2 = V0/kappa	True
3	radion on S^4: the same with the potential -(M_Pl^2/2) e^(-4 beta) 12/(r0^2 e^(2 beta)) from the curvature of the sphere, V0 = (8 pi^2/3) r0^4	True
4	n extra dimensions (T^n, n = 1..4): kinetic term (M_Pl^2/2)(n(n+2)/2)(dB)^2; the note's beta_n = sqrt(2/(n(n+2))) gives (M_Pl^2/2)(d phi)^2, not (M_Pl^2/2)(1/2)(d phi)^2; beta_n = 1/sqrt(n(n+2)) gives the 1/2	True
5	the note's eight-dimensional alpha = -1/sqrt(3), beta = 1/sqrt(12): 2 alpha + 4 beta = 0, 4 beta = 2/sqrt(3), sqrt(12) beta = 1	True
6	the kinetic coefficient solved from the reduction (n = 1..4) is n(n+2)/2	True
7	the cosmological metric with the sphere of radius b: G^A_A = torus + (6, 6, 6, 6, 3, 3, 3, 3)/b^2, R = torus - 12/b^2	True
8	radion equation on S^4: Hb' + 3 Ha Hb + 4 Hb^2 = kappa (eps/6 - pa/2 + pb/3) + Lambda/3 - 3/b^2	True
9	radion on T^4 with U: flat vacuum Lambda + kappa U0 = 0, U1 = 0; m_r^2 = kappa K_U/12	True
10	radion on S^4 with U: Lambda + kappa U0 = 6/r0^2, kappa U1 = -12/r0^2, m_r^2 = kappa K_U/12 - 2/r0^2, from V and from the 8D radion equation	True
11	S^4 with Lambda only: dS4 x S^4 (r0^2 = 9/Lambda, H^2 = 1/r0^2) solves the 00, 11, 44 equations; maximum of V with m^2 = -6 H^2, also from the 8D radion equation	True
12	S^4 with a 4-form flux: eps = rho_F, pa = -rho_F, pb = +rho_F; Minkowski needs Lambda = 9/(2 r0^2), kappa rho_F = 3/(2 r0^2); m^2 = 6/r0^2 from V and from the 8D radion equation	True
13	the note's Goldberger-Wise potential: for a static Phi(y) the Einstein-frame energy carries e^(-4 beta) (gradient part e^(-6 beta)), not e^(+4 beta)	True
14	radion coupling to a mass on the brane: -M e^(-n beta/2) = -M + (n/2) M beta; coupling sqrt(n/(2(n+2)))/M_Pl = 1/(sqrt(3) M_Pl) for n = 4; fifth force 2 (coupling M_Pl)^2 = n/(n+2) = 2/3	True

#	check	result
15	Eot-Wash: m_r > hbar c/38.6 micrometres = 5.1 meV; range of a 10 GeV radion 2.0e-17 m; gamma = v/(sqrt(3) M_Pl) = 5.8e-17	True
16	zonal harmonics on S^4 (l = 0..6): eigenvalue -l(l+3), C_l(1) = (l+1)(l+2)/2, Int (C_l/C_l(1))^2 = V/d_l; d_l = binomial(l+4,4) - binomial(l+2,4) (l <= 20)	True
17	Weyl's law on S^4: N(x)/(x^4/12) - 1 = -1/x^2 at x = 10, 100, 1000	True
18	the brane at the north pole of S^4: the embedding induces r0^2 gamma; north pole (0,0,0,0,r0); 6 of the 10 rotations of R^5 fix it (SO(4)), 4 move it	True
19	Killing spinors of S^4: nabla eps_s = s (i/(2 r0)) gt eps_s and D eps_s = (2 s i/r0) eps_s (s = +-1; residual < 1e-25 at four random points, 30 digits); Friedrich bound 4/r0^2	True
20	Dirac spectrum on S^4: Weyl's law for spinors L^4/3 versus the degeneracies 2 x 4 binomial(l+3,3) (2.0e-3 at L = 1000)	True
21	the sphere with the volume of the torus today: r = 688.9 GeV^-1; lowest levels 4.03 MeV (torus) and 2.90 MeV (sphere); dirac16complex on the sphere 2.90 MeV; levels below 5 GeV agree with Weyl	True
22	RS1 (5D, mostly minus): G^mu_nu = 3 A'' - 6 A'^2, G^y_y = -6 A'^2; bulk Lambda = -6 k^2; tensions f_UV = +6k/kappa, f_IR = -6k/kappa	True
23	Israel junction at the UV brane: K^mu_nu = -A' delta, [K^mu_nu] - delta [K] = 6k = 3 x (jump of A'), the delta part of G^mu_nu; in the signature (-,+,+,+,+) [K - g K] = -kappa S	True
24	RS1: M_Pl^2 = (1/kappa) Int e^(-2k|y|) dy over [-pi R, pi R] = (1 - e^(-2 pi k R))/(kappa k)	True
25	RS1 radion: kinetic term (3/(kappa k)) (d e^(-k pi T))^2, canonical mu = sqrt(6/(kappa k)) e^(-k pi T)	True
26	8D warped along one extra direction: G^y_y = -6 A'^2, G^z_z = 4 A'' - 10 A'^2; with only Lambda A'' = 0 and A' = 0	True
27	Goldberger-Wise: Phi'' - 4k Phi' - m^2 Phi = 0 from the bulk action, solved by e^(2ky)(C1 e^(nu k y) + C2 e^(-nu k y)), nu = sqrt(4 + m^2/k^2) = 2 + eps; V'(<mu>) = 0 at (<mu>/f)^eps = v_B/v_A, V''(<mu>) = 8 k eps^2 v_B^2 <mu>^2/f^4; m_r^2 = (4/3) kappa k^2 eps^2 v_B^2 (<mu>/f)^2 (note: 2/3)	True
28	conformally flat h = Omega^2 eta: sqrt|h| R(h) = -6 Omega box Omega, box = d0^2 - d1^2	True
29	the Higgs boson on the brane: linear terms are total derivatives; to second order L = (1/2)(dh)^2 + (1/2)(1 + 6 xi gamma^2)(dr)^2 + 6 xi gamma dh.dr - (1/2) m_h0^2 h^2 - (1/2) m_r0^2 r^2 up to total derivatives; o_2 drops out	True
30	the shift h = h' - (6 xi gamma/Z) r', r = r'/Z is canonical iff Z^2 = 1 + 6 xi gamma^2 (1 - 6 xi) (note's literal kinetic terms: 1 - 36 xi^2 gamma^2); tan 2 theta = 12 gamma xi Z m_h0^2/(m_r0^2 - m_h0^2 (Z^2 - 36 xi^2 gamma^2)) diagonalizes with (h', r') = Q(theta)(h_m, r_m); the inverse matrix needs -theta; C_h = cos - (6 xi gamma/Z) sin, C_r = sin/Z	True
31	Higgs-radion mixing in 7+1 (gamma = v/(sqrt(3) M_Pl), xi = 1, m_r0 = 10 GeV): |Z - 1| < 1e-30, |theta| = 3.5e-16, rates change by < 1e-30; warped gamma = v/(3 TeV) = 0.082	True

Numbers printed by the notebook and by the programs (largest relative difference per group, and the tolerance used by tools/verify_7p1_nb.py; integers and exact matrices must agree exactly):

quantity	values	max. relative difference	tolerance
orbit sizes of the shells $k = 1, \dots, 12$	16	0	exact
$r_4(k)$, $k = 0, \dots, 16$	17	0	exact
Weyl: $N(R)$, $R = 10, 30, 100, 300, 1000$	5	0	exact
Weyl: $N(R)/(\pi^2 R^4/2) - 1$	5	3.8×10^{-10}	1×10^{-9}
$V_\perp$, $\ell_\perp$ (GeV^{-1}, m, fm), $2\pi/\ell_\perp$	5	2.2×10^{-16}	1×10^{-12}
$\ell_\perp/r_p$, $a_0/\ell_\perp$, $R = \ell_\perp/2\pi$, M_D	5	1.7×10^{-16}	1×10^{-12}
$N(\mu)$ at 5 GeV, m_h, M_8; lattice count below 5 GeV	4	1.8×10^{-16}	1×10^{-12}
the first levels (k, r_4, $4r_4$, mass)	32	2.5×10^{-16}	1×10^{-12}
image sums $S(r)$ and Kaluza–Klein sums, $r = \frac{1}{4}, \frac{1}{2}, 1, 2$	8	1.2×10^{-15}	1×10^{-9}
$Z(5/2)$: closed form, shell sum	2	2.9×10^{-16}	1×10^{-9}
near-field and far-field ratios	5	7.8×10^{-12}	1×10^{-6}
Christoffel symbol $\{4\,00\}$ at the brane	1	0	1×10^{-15}
Γ_A as signed permutations	128	0	exact
$\mathrm{Re}(2\sqrt{2}\,U)$	256	0	exact
$\mathrm{Im}(2\sqrt{2}\,U)$	256	0	exact
example spinor: $\bar\psi\psi$, $\bar\psi\Gamma^\mu\psi$	5	0	1×10^{-15}
T_*, f_W/ρ_Λ, $1/(2\pi M_8)$	3	5.8×10^{-16}	1×10^{-12}
kinetic coefficient $n(n+2)/2$, $n = 1, \dots, 4$ (solved from the reduction)	4	0	exact
β_n^2 of the note and the canonical β_n^2, $n = 4$	2	0	1×10^{-15}
S^4 with U: $(\Lambda + \kappa U_0) r_0^2$, $\kappa U_1 r_0^2$, $(m_r^2 - \kappa K_U/12) r_0^2$	3	0	exact
S^4 with Λ only: Λr_0^2, $H^2 r_0^2$, m^2/H^2	3	0	exact

quantity	values	max. relative difference	tolerance
S^4 with a 4-form flux: Λr_0^2, $\kappa\rho_F r_0^2$, $m^2 r_0^2$	3	0	1×10^{-15}
E\"ot-Wash bound, range of a 10 GeV radion, $\gamma = v/(\sqrt{3}M_{\rm Pl})$	3	2.1×10^{-16}	1×10^{-10}
multiplicities d_l on S^4, $l = 0,\ldots,8$	9	0	exact
Weyl's law on S^4: $N/(x^4/12) - 1$ at $x = 10, 100, 1000$	3	8.2×10^{-11}	1×10^{-9}
Dirac degeneracies on S^4 and the spinor Weyl ratio	7	7.0×10^{-14}	1×10^{-9}
today: r_{S^4}, lowest masses, numbers of levels below 5 GeV	8	2.2×10^{-16}	1×10^{-12}
RS1: 3, $f_{\rm UV}\kappa/k$, $f_{\rm IR}\kappa/k$, $f^2\kappa k$	4	0	exact
strength of the warped coupling over the flat one	1	1.8×10^{-16}	1×10^{-12}
Goldberger–Wise mass coefficient: $\frac{4}{3}$ and the note's $\frac{2}{3}$	2	0	1×10^{-15}
Higgs–radion: $c = -6$, $K_{hr}/(\xi\gamma)$, $(K_{rr}-1)/(\xi\gamma^2)$	3	0	exact
Higgs–radion mixing in 7+1: $Z-1$, θ, rate changes; warped γ	5	4.4×10^{-16}	1×10^{-12}

6.3 The earlier sections are unchanged

tools/verify_7p1_nb.py compares the evaluated notebook with the backups of section 3 and with the Python programs; its lines on sections 25 and 26:

```
section 25: check outputs (all True) : 27 (True)
stage 6: without section 25 (and with the old title of the summary section) the 786 input and
    ...text cells equal those of the backup taken before section 25 was appended : True
stage 6: the 481 outputs of sections 0-24 and of the summary equal those of the backup (summary
    ...counts and tables, the private context name and measured times excepted) : True
section 25 numbers: 17 groups of printed numbers agree with tools/ref_torus_brane.py (integers
    ...and exact values exactly, floating-point numbers to 1e-15 ... 1e-6 per group) : True
section 26: check outputs (all True) : 33 (True)
stage 7: without section 26 (and with the old title of the summary section) the 815 input and
    ...text cells equal those of the backup taken before section 26 was appended : True
stage 7: the 515 outputs of sections 0-25 and of the summary equal those of the backup (summary
    ...counts and tables, the private context name and measured times excepted) : True
section 26 numbers: 15 groups of printed numbers agree with tools/ref_radion.py (integers and
    ...exact values exactly, floating-point numbers to 1e-15 ... 1e-9 per group) : True
VERDICT: ALL CROSS-CHECKS PASSED
```

6.4 Files

SHA-256 of the files (at the time this document was generated):

```
7+1.nb                          41d252f9852f210c228f45e1ac956d2d7ef26faeb0062a5880280e1aae0ef
    ...fc7
tools/make_7p1_nb.py            996d26c9552cd44902560af16a1683fa71457261f18fefaeddae87bd99a2b
    ...144
tools/nb_part_gravity.py        c4caf2ad5c932e8f603341edade5f4b98fa784feb05906c051eab7c884cd6
    ...8f8
tools/nb_part_ks_gravity.py     a656c0ad040b4576b75b17e029545109e531c8e656ad99bfdb5cba946e967
    ...772
tools/nb_part_sm.py             43e92beed57453a4b57931de6d603ac6c218efba6e4005133b69b8054a975
    ...c5b
tools/nb_part_coupled.py        e630521d64132e71afc861114f699eb06454557cf14a57ad8ec1eef96247a
    ...ae0
tools/nb_part_dark.py           10ad8fb8ae25fa07bbee758c653995acd4e2c80733059a8b3648bd1593e46
    ...db0
tools/nb_part_torus_brane.py    fb904ff8d85b904c55b3b2a8f60fb0cb047f1ccf1b98db3364092f2bae0b2
    ...162
tools/nb_part_radion.py         53c5604707a3e22228c840ee0a2d2125616f9410f75a5535041dfaf18f497
    ...c00
```

```
tools/lovelock71.py                d93ecbd0f82adb972218d59dfe32999a4af944945ca9758607f3a633e1af9
    ...5a1
tools/ref_gravity.py               f07cf8adaff585af89c87d19532f1aaa7db606ad88ae90483b2d1f41341ed
    ...f36
tools/ref_ks_gravity.py            b6488ef24b1c16aacd8ff6cf6a5a1ba3f316e082449ad97480022d1a26383
    ...27f
tools/ref_sm.py                    f3f80806893f42bbf166e1f1de896ffd611dbfe847685da8e3ee367e9cde6
    ...aba
tools/ref_coupled.py               695dbd9375753653472d9c37502f0067c4a64fe01a64ba8c05e8b2c5da85a
    ...a58
tools/ref_dark.py                  1600e6b57ab0383b4f2b664e2796d3864c9afa77a9ed59f007f66e1139453
    ...ff5
tools/ref_torus_brane.py           1f623656ab0f5c8ee43cb310f7bfd580cbba5a7385da2963adaae4d2f8bda
    ...af7
tools/ref_radion.py                e9d47176b9f2d2232a84b3b82188bd3b0a19cde3dbe1ba013c1fe5abff634
    ...1ce
tools/ref71.py                     1a7a06eb0add3f459611776dc9c474a5bb959d947a9db949db4612b1ab943
    ...4e0
tools/ref44.py                     ab100c3ebcc055fbe59c132a02f80ecb17630e934e99d59f88e919ec8064f
    ...af5
```